The Collapse of Heterogeneity in Silicon Philosophers

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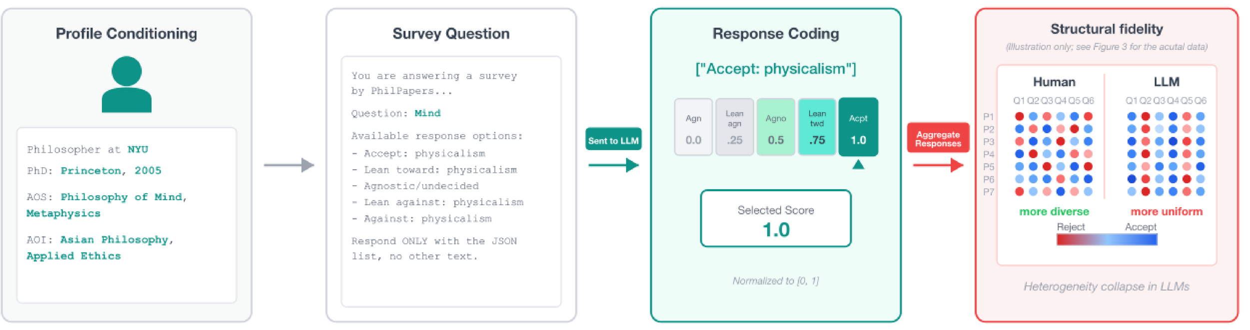
Question

Can an LLM stand in for a population of minds?

Setup

- $N = 277$ PhilPeople philosophers, 100 PhilPapers 2020 questions
- 7 profile-conditioned LLMs, temperature 0
- Accept $\to$ Reject coded onto $[0, 1]$
- Checked against PhilPapers 2020 ($N = 1{,}785$)

Silicon sampling



How we measure spread

- **Resp%** is how often a source answers a question. Humans skip more; models answer more.
- **Per-Q Var** is the average variance of answers *on the same question*. High variance means philosophers actually disagree.

Source	Resp%	Per-Q Var
Human	38.9%	0.053
Claude-Sonnet-4.5	60.0%	0.026
Llama-3.1-8B	69.1%	0.026
Mistral-7B	67.7%	0.028
Llama-3.1-8B (FT)	66.5%	0.019
Qwen-3-4B	83.3%	0.016
GPT-4o	60.2%	0.014
GPT-5.1	59.9%	0.014

Humans disagree more than every silicon sample. Models fill in more answers, with less spread.

Normative domains fail most

- Models look accurate where philosophers already agree (aesthetics, language).
- They fail on applied ethics and philosophy of religion — the domains with the most human disagreement, and the ones that matter for alignment.
- Prediction error tracks human variance ($r = 0.75$): the model is not "better at ethics"; it just benefits from consensus questions.

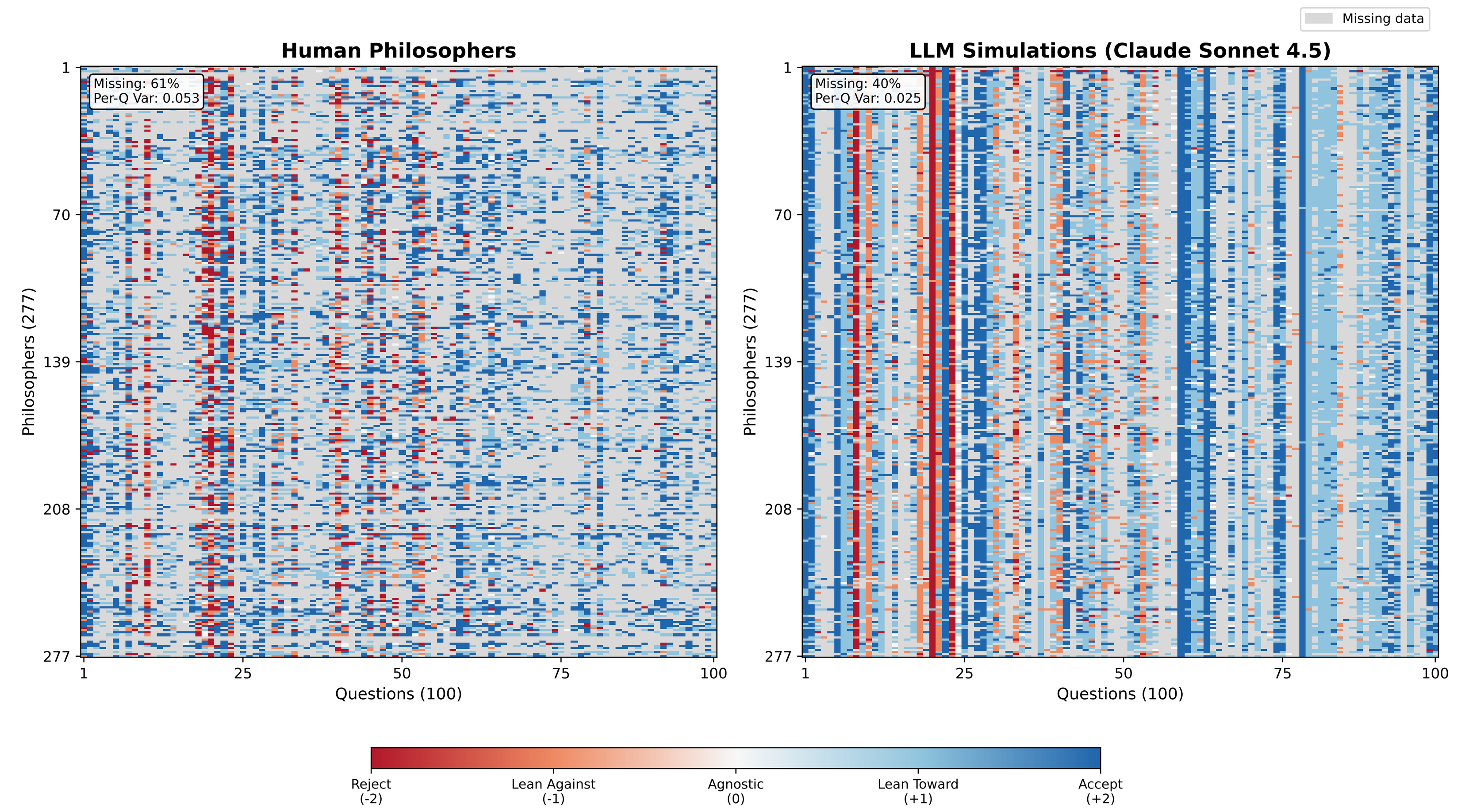
WM@Booth 2026

A social world model is adequate only if the distribution it induces matches the distribution of the population it represents.

Silicon philosophers **fail this test**: they are **too uniform**, and they disagree along the **wrong axes**.

Take a picture for the paper and data explorer.

Human vs. Claude



Each cell is one philosopher on one question. Humans are speckled; Claude forms vertical bands.

PCA: the axes of disagreement

PCA finds the main directions along which philosophers differ.

- Humans: five independent axes. PC1 is naturalism / meta-ethics, not a left–right spectrum.
- LLMs collapse onto fewer, topical axes (mind uploading, spacetime) instead of that cross-domain pattern.
- Component loadings do not match humans. Even the best alignment stays weak.

KL, JS, and Mantel

- **KL** / **JS**: distance of the answer mix from humans (lower is closer). Claude is nearest (0.052 / 0.010); GPT-4o is farthest (1.012 / 0.145).
- **Mantel**: permutation test on the 100×100 question-correlation matrix — is the structure closer than chance?
- Claude and both GPTs pass Mantel ($p < 0.01$). Llama-FT and Mistral are weaker ($p < 0.05$). Llama base and Qwen do not.

Invented specialist links

Each silicon philosopher is prompted with a PhilPeople profile, including areas of specialization and areas of interest. Models treat those labels as a cue for the *question's topic*, and invent a stance that "fits" the words.

- Philosophy of Biology in the prompt $\to$ the "biological" view of personal identity on the test item, because the words overlap. Real biology specialists do not show this link.
- Ancient philosophy in the prompt $\to$ "must be Aristotelian" on practical reason. Humans: no such link.
- The same lexical shortcut runs the other way: biology specialists are pushed *against* psychological identity.
- These are bogus relationships: large and significant in the silicon sample, absent in the 277-philosopher ground truth and in PhilPapers 2020.

	Human		LLM	
Specialist effect	Diff	Sig	Diff	Sig
Biology $\to$ biol. identity	$+11.4$ pp	n.s.	$+43$ pp	4/7
Biology $\to$ psych. identity	$+4.1$ pp	n.s.	-65.7 pp	3/7
Ancient $\to$ Aristotelian	$+1.9$ pp	n.s.	$+68.9$ pp	7/7

DPO fine-tuning

DPO on Llama-3.1-8B improves structure, not diversity.

- Question-to-question correlation doubles, but still far from human.
- The answer mix drifts: 34.7% Agnostic vs. 1.4% in humans.
- Human-like spread is not restored.